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CGRemote™

Protocol Specifications
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1 How to use CGRemote protocol

CGRemote™ is an advanced remote control interface that allows to control CGPlayout™ through a set of commands. CGRemote™ interacts with CGPlayout managing in easy way all its features: it is able to load and play graphics templates created by MoreCG™ or CastaliaCG™, and also to setup and play each ClassX MediaPlugs™.
CGRemote™ is a middle level application designed to integrate CGPlayout™ with the more commons TV Automations through a TCP/IP connection.

CGRemote™ interacts with MoreCG™, CastaliaCG™, CoralCG™ graphics templates through their filename (<gfxprojectpath>) or their ID into CGPlayout list. Each template is identified uniquely by a string 'ID' that will be used as a unique key to detect the template and interact with it. The ID is not case sensitive and cannot be null (i.e. “”). Any leading or trailing white spaces are removed automatically by the CGRemote protocol handler. The ID may contain only [a..z],[A..Z][0..9]["-","_"] characters.
Before call any commands regarding a graphic template it is mandatory to select it. This operation is implemented by SelectProjectByID <ID> or SelectProject <gfxprojectpath> commands.
CGPlayout allows to play a video and audio files into a special element usually named “Video/Audio” content. It is also possible to play audio and video files synchronously with a graphic template.

Following the same way, each MediaPlugs™ is identified uniquely by its position into the list. The command GetAvailablePlugins return a formatted list that contains the available plugIns and their index.

1.1 Connecting to CGRemote

The users can interact with CGRemote through a TCP/IP port (the COMMUNICATION_PORT, default is 101). CGRemote processes each command sequentially and always returns either the command execution result or an error code in case the command fails. Each command is processed immediately.

Users can set their preferred ports in the CGRemote preference panel.

1.2 Special characters

All sent and received protocol strings are in ASCII format and are always closed by quote chars. They may contain special chars in ANSI-C format (escape sequences) as line feed, carriage return etc. The following table contains all supported special characters:

NEWLINE	\n
HORIZONTAL TAB	\t
BACKSPACE	\b
RETURN	\r
FORM FEED	\f
BACKSLASH	\\
SINGLE QUOTE	\’
DOUBLE QUOTE	\"

Example:
>**GETPROJECTOBJECTS**
<OBJECTNAME="local",VALUE="Restaurant \"The Boat\" Milan",KIND=TEXT

In this case the **value** of the **local** object is **Restaurant "The Boat" Milan**

In case of file path string this protocol supports both Back Slash (DOS/Windows) and Slash (Unix/Linux/Mac) characters.

NOTE: The protocol is always parsed line by line and each line sent to or received from the protocol handler is always terminated by a **newline** character. It is always possibile to include multiple newlines characters in a command line by replacing any occurrence with the "\n" sequence.

>**GETPROJECTOBJECTS**
<OBJECTNAME="local",VALUE="Restaurant\nMilan",KIND=TEXT

the VALUE will be parsed as

1.3 Commands returning lists

Some of the commands in the protocol can return a list of values. The lists are sent to the calling application by a sequence of Line-feed ('\r') terminated lines.

1.4 Project identification

Each project is identified uniquely by a string 'ID' that will be used as a unique key to detect the project and interact with it. The project ID is not case sensitive and cannot be null (i.e. "”). Any leading or trailing white spaces are removed automatically by the CGRemote protocol handler. The project ID may contain only [a..z], [A..Z][0..9]["-","_"] characters.

NOTE: sometimes in this documentation we refer to Project or Engine names, they both indicate the same things.

1.5 The Commands syntax parsing

The TCP/IP protocol parser is based on the ArgParser by ClassX, a specialized set of classes that allows flexible parsing of the commands sent to the protocol handler.

The ArgParser will parse the built-in command templates for each command of the protocol.

The **Help** command gives a list of the entire set of commands allowed by the protocol. Giving a closer look at the list, we can see that almost any command is followed by a series of parameters. Each parameter is marked with a control character sequence that rules the interpretation of the command:

PARAMETER Marker	Interpretation
<no sequence>	Normal parameter
\k	Keyword parameter
\r	Required parameter
\s	Swithch parameter (TRUE or FALSE)
[...]	Param information: ignored when parsing

The ArgParser will process each command by the following algoritm:

1. Discard param information like square bracket suggestions.
2. Search and assign the Required parameters. If one or more Required parameters is missing, then there will be a parsing error.
3. Search and assign the Keyword parameters.
4. Search and assign the Switch parameters.
5. Assign the remaining arguments to the Normal parameters.

An example: the following command template

```
"COMMAND/r/k,PARAM0/r,PARAM1[A|B|C]/r,PARAM2/k,SWITCH/s"
```

the above means that

```
COMMAND is a keyword and is required
PARAM0,PARAM1 are required but not a keyword
PARAM2 is a keyword
SWITCH is a true/false switch
```

will parse the following argument string

```
COMMAND=comando PARAM0=-123.453, \"PIPPO \\\"LA\\\" SCOPA\", PARAM2=\"This is Param2\" , SWITCH=TRUE"
```

as follows:

```
COMMAND = comando
PARAM0 = -123.453
PARAM1 = PIPPO "LA" SCOPA
PARAM2 = This is PARAM2
SWITCH = true
```

Note that arguments are normally a comma separated list, but also spaces are allowed. Also complex control char sequences are supported. Named arguments always have the precedence in the value assignment:

```
"ARG0/k,ARG1\k"
```

will parse the following argument string

```
"ARG1=ciao,ARG0=pippo"
```

as follows:

ARG0=pippo
ARG1=ciao

and will parse the following argument string

```
"ciao,pippo"
```

as follows:

```
ARG0=ciao
ARG1=pippo
```

More generally the sequence of the arguments is always intepreted left to right, except that "named" arguments (keywords) have the precedence.

The **Switch** argument must be specified by assigning its name to a "true" or a "false" value. It can also be specified by simply passing the argument name:

"DISPLAY\s\k" will parse the argument "DISPLAY=FALSE" as false, "DISPLAY=TRUE" as true, "DISPLAY" as true, "" as false.

1.6 Note

ATTENTION: All obsolete commands are tagged as **{DEPRECATED}**. These commands may be not supported by future protocol releases. Please avoid to use them.

1.7 Application commands

Help

Return the list of the supported commands.

Log <source, text>

Write a text into the LOG file. This may be usefull to add comments into the LOG file. Source tell who writes the comment

Load <cgp project path> [APPEND] [IGNORETEMPLATES] [IGNOREPLUGINS]

Load the CGPlayout project. This operation may require few seconds due to the graphics elements initialization.
APPEND is used to append the project to the current; IGNORETAMPLETES to ignore the templates of the new project; IGNOREPLUGINS to ignore the plugins of the new project.
The default settings for these arguments are: APPEND = false; IGNORETEMPLATES = false; IGNOREPLUGINS = false.
Return "Ok" if the operation has finished without errors or one of these errors types: ERR_LOADPROJECT, ERR_INVALIDFILENAME, ERR_BADPARAMETERS

Example:

```
-> load C:\ClassX_Applications\cgplayout\project\news\notiziario_01.cgp
<- OK

-> load C:\ClassX_Applications\cgplayout\project\news\notiziario_01.cgp APPEND
<- OK

-> load C:\ClassX_Applications\cgplayout\project\news\notiziario_01.cgp IGNOREPLUGINS
<- OK
```

Quit <saveproject>

Save the current project and then close the application.

Clear

Stop all playing graphics templates and MediaPlugs and then unload the CGPlayout project to release memory resources.

SetPlayMode [AUTO|LIVE]

Select the playmode between AUTO or LIVE

GetPlayMode

Return the current playmode [AUTO|LIVE]

SetOnAir [ON|OFF]

Enable the OnAir properties of CGPlayout

GetOnAir

Return the current playmode [ON|OFF]

GetStatus [PLAYER|LOADER]

Return the current status of PLAYER or LOADER modes. CGPlayout switches the mode automatically: the default status is PLAYER, and it is LOADED when CGPlayout is loading a project.
The command GETSTATUS PLAYER returns:

- READY, when none project is playing.
- PLAYING PROJECT+ projectName + AT timecode, when a project is playing.

The command GETSTATUS LOADER returns:

- READY, when none project is playing, and the loadProject command can be executed.
- NOT READY when a project is playing.

GetLoadedProjects

Return a formatted list of the loaded projects. Each line ends with a '\r' (Line feed o Return) character.

Example:

```
<list_pos_id>,<gfxprojectpath>,<audiopath>,<videopath>
"0","c:\\xxx\\yy\\news.mcg"
"NEWS","c:\\xxx\\yy\\news.ccg"
"1","c:\\xxx\\qq\\news.ccg","audio.wav","video.avi"        // graphic + audio + video
"2","c:\\xxx\\zz\\news.mcg",,"video.avi"                  // graphic + video
"3","c:\\xxx\\zz\\news.mcg","pippo.wav"                   // graphic + audio
"4",,"pippo.wav"                                          // only audio
```

Each project has a image preview icon (90x72 pixels) in TGA format which path can be get simply replacing the ".tga" extension into the project path.

Example: "c:/xxx/yy/news.mcg" icon -> "c:/xxx/yy/news.tga".

GetInfo <PROJECTFILENAME | RESOLUTION, PROJECTDIR>

Return the project information: the file name of the loaded file and the resolution of the video output.

The command result depends by the required property.

PROJECTFILENAME: project file name

```
->GETINFO PROJECTFILENAME
<-C:\ClassX_Applications\cgplayout_oem\example.cgp
```

RESOLUTION: the project resolution

```
-> GETINFO RESOLUTION
<- PAL (720x576) FPS:25.0 Aspect:PAL 4:3 DV Interlaced
```

PROJECTDIR: projects folder name reading it from application preference

```
->GETINFO PROJECTDIR
<-C:\ClassX_Applications\cgplayout\project
```

RestartGFBServer

Stop and restart the ClassX Graphic Server.

ReclaimMemory

Free memory resources.
WARNING: calling this command while graphics elements are playing may cause drop frames.

1.8 Graphics templates commands

AddProject <GFXPROJECTPATH, PROJECTID>

Add a MoreCG or CastaliaCG graphic template into CGPlayout at the last position list.

The template objects are not updated to the currently DataSource values. In this case we suggest to call the update dataSources commands to update the template objects.

Example:

```
-> AddProject "C:\\\\ClassX_Applications\\\\morecg\\\\project\\\\news\\\\SD_notizia.mcg","TG
NEWS"
<- OK
```

DelProject <GFXPROJECTPATH>

Remove the graphic template from CGPlayout list identifying it by the given <GFXPROJECTPATH>

Example.:

```
-> DelProject "C:\\\\ClassX_Applications\\\\morecg\\\\project\\\\news\\\\SD_notizia.mcg"
<- OK
```

DelProjectByID <POSID>

Remove a graphic template from CGPlayout list identifying it by the given project ID <POSID>

SelectProjectByID <POSID>

Select the current graphic template to manage at the given project ID <POSID>

SelectProject <GFXPROJECTPATH>

Select the current graphic template to manage identifying it by the given <GFXPROJECTPATH>

SetProjectSpeed <SPEEDPERC>

Set the playing speed in percentage [1..500]% of the selected graphic template

SetProjectLoops <LOOPS>

Set the loops number of the selected graphic template. Use -1 for infinite loops.

AddMediaProject <VIDEOFILE=videomediafile [AUDIOFILE=audiomediafile]>

Add special "media" element into CGPlayout usually named "Video/Audio". If the video or audio file cannot be load this operation will be rejected.

Example:

```
-> AddMediaProject VIDEOFILE="C:\\\\ClassX_Applications\\\\shared\\\\video\\\\video.avi"
<- OK

-> AddMediaProject AUDIOFILE=" C:\\\\audio\\\\tg.wav"
<- OK

-> AddMediaProject VIDEOFILE="C:\\\\video\\\\video.avi" AUDIOFILE=" C:\\\\audio\\\\tg.wav"
<- OK
```

SetProjectMediaFile <VIDEOFILE=videomediafile [AUDIOFILE=audiomediafile]>

Assign video and audio content to the current selected graphic template

SetProjectMediaProperties <[
VIDEOFILE_ALPHAENABLE=[true | false] VIDEOFILE_ALPHAFILLED=[true | false]
VIDEOFILE_AUDIODELAY=[0..4000]
VIDEOFILE_AUDIOONLY=[true | false]
VIDEOFILE_FIELDSPOLARITY=[EVEN | ODD | NONE]
VIDEOFILE_LOOP=[true | false]
VIDEOFILE_SCALEHQ=[true | false]
VIDEOFILE_SCALETOFIT=[true | false]
VIDEOFILE_SYNCHTOVIDEOREFRESH=[true | false]
VIDEOFILE_VOLUME=[0..255]
AUDIOFILE_LOOP=[true | false]
]>

Set one or more than one Video and Audio properties.

```
Es.:
-> SetProjectMediaProperties VIDEOFILE_AUDIODELAY=300
<- OK

-> SetProjectMediaProperties VIDEOFILE_AUDIODELAY=300 AUDIOFILE_LOOP=true
<- OK

-> SetProjectMediaProperties VIDEOFILE_SCALETOFIT=true
<- ERR_MEDIAFILENOTFOUND
```

UpdateDataSources

Force to reload the DataSource file of the current graphic template.

ReloadProjectDataSourcesByID <PROJECTID>

Force to reload the DataSource file of the graphic template at the given project ID PROJECTID.

ReloadProjectDataSources <PROJECTNAME>

Force to reload the DataSource file of the graphic template at the given index PROJECTNAME.

Start

In LIVE playmode this command plays the selected graphic template or resume it if it is paused
In AUTO playmode this command plays all graphics templates of CGPlayout starting from the first element of the list.

Stop

Stop the current playing graphic template.

Take [<frame>,<takemode[SINGLE|START_SEQUENCE|END_SEQUENCE]>]

Make a take (screenshot) of the given frame of the selected graphic template, clearing the background before if clear is true. If frame is -1 CGPlayout will use the template posterframe. A sequence of take commands will compose (blend) different templates frames into a single buffer.

This supports different takemodes:

- **SINGLE** clears the buffer, takes a screenshot and sends the image to the output device.
- **START_SEQUENCE** clears the buffer, takes a screenshot but does not send the image to the output device. It means that we have started a sequence of take commands.

- **END_SEQUENCE** ends the take sequence and sends the image to the output device.

TakeAtRatio [<currenttime>,<totaltime>,<takemode[SINGLE|START_SEQUENCE|END_SEQUENCE]>]

Normally the **Take** command works in a frame range defined between 0 and the template total frames. This special Take command computes a ratio value using the formula <currenttime>/<totaltime>. This value is clamped between [0,1] where 0 is the template start, and 1 is the template end. The rendered frame is computed by the timeline interpolation at the given ratio.
The <takemode> param is the same of the Take command.

OpenProjectEditor

Open the ClassX Application that is able to edit the current graphic template. Be careful to use this command because when a new application is launched it requires resources, and it may cause drop frames on the ouptut.

1.9 Editable objects of a graphic template

All of the following commands interact only with the selected graphic template.

GetProjectObjects

Return a formatted list of all editable objects contained in the MoreCG or CastaliaCG graphic template. The list has the following format:

OBJECTNAME=<name>,VALUE=<value>,KIND=[TEXT]

The object KIND describes the object type as text, image, video etc.

Return

```
OBJECTNAME="name",VALUE="Cristian",KIND=TEXT
OBJECTNAME="surname",VALUE="Smith",KIND=TEXT
OBJECTNAME="team",VALUE="gfx/milan.tga",KIND=IMAGE
```

GetProjectInfo <TOTFRAMES | NOBJECTS | CURRFRAME | DESCRIPTION | AUDIOFILE | VIDEOFILE | GFXFILE | ICONFILE|FRAMEMARKER|POSTERFRAME|FPS|RESOLUTION>

Return the graphic template information as total frames, number of objects, the current frame, the template description, the file name, the template icon file path, the pause frames, the poster frame, the template FPS and resolution.

The command result depends by the required property.

TOTFRAMES: template total frames

```
->GETPROJECTINFO TOTFRAMES
<-250
```

NOBJECTS: project editable objects number

```
->GETPROJECTINFO NOBJECTS
<-5
```

CURRFRAME: the current drawn frame

```
->GETPROJECTINFO CURRFRAME
<-127
```

DESCRIPTION: the project description

```
->GETPROJECTINFO DESCRIPTION
<-"This project has been designed to display dynamic news"
```

AUDIOFILE: the audio file path

```
->GETPROJECTINFO AUDIOFILE
<-"C:\\audio\\tg.wav"
```

VIDEOFILE: the video file path

```
->GETPROJECTINFO VIDEOFILE
<-"C:\\audio\\video.avi"
```

FRAMEMARKER: a list of frame markers returning for each elements the frame and the marker type.

```
>GETPROJECTINFO FRAMEMARKER
<FRAME=10,TYPE="TRIGGER_AND_PAUSE"
 FRAME=40,TYPE="TRIGGER_AND_LOOP_START"
 FRAME=70,TYPE="TRIGGER_AND_CONTINUE"
 FRAME=100,TYPE="TRIGGER_AND_PAUSE"
 FRAME=110,TYPE="TRIGGER_AND_LOOP_END"
```

GFXFILE: the graphic template file path

```
-> GETPROJECTINFO GFXFILE
```

```
<-"C:\\\\ClassX_Applications\\\\morecg\\\\project\\\\news\\\\SD_notizia.mcg"
```

ICONFILE: the project icon file path. The icon is a TGA image with a low resolution (90x72) pixels

```
-> GETPROJECTINFO ICONFILE
<- "C:\\\\ClassX_Applications\\\\morecg\\\\project\\\\news\\\\SD_notizia.tga"
```

POSTERFRAME: the project posterframe

```
-> GETPROJECTINFO POSTERFRAME
<- 86
```

FPS: the project frame per second

```
-> GETPROJECTINFO FPS
<- 25
```

RESOLUTION: the project resolution

```
-> GETPROJECTINFO RESOLUTION
<- WIDTH=720,HEIGHT=576
```

EditProjectObject <OBJECTNAME>,<VALUE>

Allows to modify the value of all editable objects with the given name contained into the graphic template. This command can be applied to all objects kind and the result depends on their implementation. For example this command changes the text value for a TEXT object and also set the file path for an IMAGE or VIDEO object.

Example

```
-> EDITTEMPLATEOBJECT "name","Adam"
<- OK

-> EDITPROJECTOBJECT "team","gfx/milan.tga"
<- OK

-> EDITPROJECTOBJECT "logo","gfx/milan_logo.avi"
<- OK
```

1.10 MediaPlugs commands

The specific commands of each plugIn are explained into the MediaPlugs_PROTOCOL_ENG document.

GetAvailablePlugins

Return a formatted list that contains the index of each MediaPlugs

```
-> GetAvailablePlugins
<- 0,Clock
<- 1,DigiClock
<- 2,Logo
<- 3,Ticker Horizontal
<- 4,SMSTicker
<- 5,Ticker Horizontal
<- 6,MediaText
```

GetPluginStatus <PLUGINID>

Return the status of the MediaPlug at the given index <PLUGINID>

RUNNING	->	the plugIn is playing
SUSPENDED	->	the plugIn is suspended
STOPPED	->	the plugIn is stopped

Example: return the status of the plugin at position 3.

```
-> GetPluginStatus 3
```

```
<- RUNNING
```

SendPluginCommand <PLUGINID>,<PLUGINCOMMAND>

Send a command to the plugin with the given ID.
Each plugin can have a set of custom commands to control many internal aspects, behaviors and settings. Normally only the MediaPlugs expose this feature. For more information please consult the **MediaPlugs Commands** section.

To obtain a correct interpretation of the command parameter, it should be written by replacing special characters with the equivalent ANSI-C escape sequence. See the **Special Characters** table.

Example: change the image file of a Logo plugin.

```
-> SendPluginCommand 2, "setFileName \"C:\\Picture\\classx.tga\""
<- Ok
```

StartPlugin <pluginID>

Play the MediaPlug at the given index <PLUGINID>

StopPlugin <pluginID>

Stop the MediaPlug at the given index <PLUGINID>

StopAllPlugins

Stop all playing MediaPlugs

SetPluginBounds <pluginID>,<X>,<Y>,<WIDTH>,<HEIGHT>

Set at the same time the plugIn position and size

WaitPluginStatus <pluginID>, <RUNNING | SUSPENDED | STOPPED>, [<timeout>]

Wait until the plugin reaches the desired state. Return Ok if the plugin has reached the desired state, ERR_TIMEOUT otherwise. The timeout default value is 3000 milliseconds. Set the timeout value to change the default value.

Example:
wait the STOPPED state of the plugin at the position 3.

```
-> WaitPluginStatus 3, STOPPED
<- OK
```

Example:
wait the RUNNING state of the plugin at the position 3, with 5 seconds of timeout.

```
-> WaitPluginStatus 3, RUNNING, 5000
<- ERR_TIMEOUT
```

2 MediaPlug commands

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PlugIns™ commands list

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2.1 Common commands

Each plugIn supports a set of remote commands that allows to change its appearance, position and settings. The plugIns have some common properties as position, dimension described in this section, while others properties depend from their implementation and will be illustrated into the related chapters.

SETPOSITION <X>,<Y>

Move the plugIn to the screen location X,Y. The parameter with value equals to -1 will be ignored.

Example: move the content to X = 100 coordinate keeping the current Y coordinate.

```
>SETPOSITION 100,-1;
<OK
```

SETSIZE <WIDTH>,<HEIGHT>

Resize the plugIn to the given WIDTH and HEIGHT. The parameter with value equals to -1 will be ignored.

Example: set the content width to 100 keeping the current height.

```
>SETSIZE 100,-1;
<OK
```

{DEPRECATED} SETVISUALPRIORITY <VALUE>

Set the plugIn visual priority in a range value [0,10000]: 0 is the lower layer and 10000 is the higher layer.

NOTE: This command does not produce any result.

Opening "**Remote Control Tester**" in ClassX distribution and **connecting** to LiveBoard, it is possible to view the list of available commands for each Plugin.

Just insert this command in the "**Command**" area:

LBC_GETCONTENTCOMMANDS <value>

2.2 Logo

SETFILENAME <FILE>

Set logo file name.

Example
```
>SetFileName \"C:\\\\Pictures\\\\image.png\"
<OK
```

SETLOGOPROPERTY <PLAYSPEED>, <PLAYMODE>, <FIELDFLAGS>, <USEALPHA>, <FPS>

Set the Logo properties. You can set one or more properties in the same command:

PLAYSPEED speed value percentage over original speed, used to play the logo [0..200]. The default is 100 (the original speed);

PLAYMODE mode used to play the logo [0..9]. The default value is 3 [PLAY_LOOP].

- 0 = still first frame [PLAY_STILLFIRST];
- 1 = still last frame [PLAY_STILLLAST];
- 2 = play one time only [PLAY_ONCE];
- 3 = loop animation [PLAY_LOOP];
- 4 = reverse loop animation [PLAY_REVLOOP];
- 5 = ping-pong animation [PLAY_PINGPONG];
- 6 = play one time only automatically [PLAY_ONCE_AUTO];
- 7 = loop animation automatically [PLAY_LOOP_AUTO];
- 8 = reverse loop animation automatically [PLAY_REVLOOP_AUTO];
- 9 = ping-pong animation automatically [PLAY_PINGPONG_AUTO];

FIELDFLAGS field handling [0 = keep original interlace fields, 1 = de-interlace frame];

USEALPHA use the alpha information of the logo if present (default is TRUE);

FPS play speed of the animation expressed in frames per second.

Example:
```
>SetLogoPropery PLAYSPEED=50
<OK
```
Set the play speed of the Logo to the 50% of the normal speed.

```
> SetLogoPropery PLAYMODE=5,USEALPHA
<OK
```
Ping-pong animation and maintain the frames alpha value.

2.3 Timer

SETSTARTTIME <VALUE>

Set the Timer start time in seconds or milliseconds. To set it in seconds it is sufficient to give it a numeric value, or the same value followed by “s”, while to set it in milliseconds the numeric value must be followed by “m”.

Example 1: set the start time to 20 seconds.

```
>SetStartTime \"20\"
<OK
```

Example 2: set the start time to 20 seconds.

```
>SetStartTime \"20s\"
<OK
```

Example 3: set the start time to 20000 milliseconds.

```
>SetStartTime \"20m\"
<OKSETENDTIME <VALUE>
```

Set the Timer end time in seconds.

Example: set the end time to 60 seconds.

```
>SetEndTime 60
<OK
```

SETTIMERDIRECTION <[UP|DOWN]>

Set the timer direction: set UP for a timer and DOWN for a countdown

Example: setup the timer for a countdown

```
>SetTimerDirection DOWN
<OK
```

RESETTIMER

Reset the timer to start time seconds.

STARTTIMER

Start the timer count.

STOPTIMER

Stop the timer count

GETTIMEMILLIS

Return the current timer time in milliseconds

SETTIMERFORMAT

Set the time format of the timer. The only supported formats are the following:

- `"s:tt"` - seconds and milliseconds.
- `"ss:tt"` - seconds and milliseconds with 2 cyphers for the seconds always visible.
- `"s"` - seconds.
- `"ss"` - seconds with 2 cyphers always visible.
- `"sss"` -seconds with 3 cyphers always visible.
- `"m:ss"` - minutes and seconds.
- `"mm:ss"` - minutes and seconds with 2 cyphers for the minutes always visible.
- `"mmm:ss"` - minutes and seconds with 3 cyphers for the minutes always visible.

- "HH:mm:ss" - hours, minutes and seconds.
- "HH:mm" - hours and minutes.
- "HH:mm:ss:tt" - hours, minutes, seconds and tenths of second.

Example: setup the timer to display hours, minutes, seconds and tenths of second.

```
>SetTimerFormat \"HH:mm:ss:tt\"
<OK
```

2.4 Horizontal and Styled Ticker

ADDTICKERTEXT <TEXT>, <LOOPS>

Set the text to scroll into the Ticker defining also the loops numbers. Use –1 for infinite loop.

Example: set the text and loops to 3 times

```
>AddTickerText \"Tonight at 21.30: Music\", 3
<OK
```

CLEARPLAYLIST [<IMMEDIATELY>]

Clear the Ticker files playlist. If IMMEDIATELY is true the Ticker execution will be also interrupted.

Example:

```
>ClearPlaylist
<OK

>ClearPlaylist IMMEDIATELY=true
<OK
```

ADDPLAYLISTFILE <FILEPATH,MAXRECORDS_NUM,MAXRECORDS_FROMLAST>

Add the given file to the playlist at the last position. The file path may refer to either local or remote URL file. If MAXRECORDS_NUM is greater than 0, limit the number of records imported from the file. In this case MAXRECORDS_FROMLAST defines where the limit is applied: from the first or the last records.

Example:

```
>AddPlayListFile \"C://Datasource//tg.txt\"
<OK

>AddPlayListFile \"http://www.classx.it//prova.txt\"
<OK
```

ADDPLAYLISTFILES <FILEPATHS=<FILEPATH, MAXRECORDS_NUM, MAXRECORDS_FROMLAST>>

Add the given files to the playlist at the last position. The file path may refer to either local or remote URL file. If MAXRECORDS_NUM is greater than 0, limit the number of records imported from each file. In this case MAXRECORDS_FROMLAST defines where the limit is applied: from the first or the last records.

Example:

```
>AddPlaylistFiles
FILEPATHS=\"FILEPATH=\\\"C:\\\\\\\\\ClassX_Applications_v6\\\\\\\\shared\\\\\\\\datasource\\\\\\
\\class_cnbc\\\\\\\\tickers\\\\\\\\ita\\\\\\\\DAX30_INDEX\\\\\\\\2268719894846_Adidas
Ag.rss\\\"\\r\""
<RCVD: Ok
```

RESETPLAYLIST

Reset the playlist execution forcing the Ticker to restart by the first playlist file.

Example:

```
>ResetPlaylist
<OK
```

GETCURRENTTEXTDURATION

Return the scrolling time in milliseconds of the current text. The duration depends by the device resolution, the ticker width and the font size. Call this command while the text is scrolling.

Example:

```
>GetCurrentTextDuration
<21000
```

SETTICKERPROPERTY <SPEED> , <TEXTUPPERCASE>, <TEXTSEPARATOR>, <AUTOREMOVE>

Set the ticker properties. You can set one or more properties in the same command:

SPEED the ticker speed (px/s) [1, 30];

TEXTUPPERCASE true if convert to uppercase the ticker text;

TEXTSEPARATOR the ticker text separator;

AUTOREMOVE stop the ticker on finish the playlist execution;

Examples:

```
>SETSTYLEDTICKERPROPERTY "SPEED=8"
<OK

>SETSTYLEDTICKERPROPERTY "TEXTUPPERCASE=TRUE"
<OK

>SETSTYLEDTICKERPROPERTY "TEXTSEPARATOR=\" ~ \""
<OK

>SETSTYLEDTICKERPROPERTY "AUTOREMOVE=\"TRUE\""
<OK
```

2.5 Styled Ticker Horizontal

SETSTYLEDTICKERPROPERTY <SPEED> , <TEXTUPPERCASE>, <TEXTSEPARATOR>, <AUTOREMOVE>, <TEXTDIRECTIONMODE>

Set the ticker properties. You can set one or more properties in the same command:

SPEED the ticker speed (px/s) [1, 30];

TEXTUPPERCASE true if convert to uppercase the ticker text;

TEXTSEPARATOR the ticker text separator;

AUTOREMOVE stop the ticker on finish the playlist execution;

TEXTDIRECTIONMODE the ticker text direction [AUTOMATIC | LEFT_TO_RIGHT | RIGHT_TO_LEFT];

Note. AUTOMATIC: Automatically detect the text direction LTR or RTL

Examples:

```
>SETSTYLEDTICKERPROPERTY "TEXTDIRECTIONMODE=AUTOMATIC"
<OK

>SETSTYLEDTICKERPROPERTY "TEXTDIRECTIONMODE=LEFT_TO_RIGHT"
<OK

>SETSTYLEDTICKERPROPERTY "TEXTDIRECTIONMODE=RIGHT_TO_LEFT"
<OK
```

2.6 MediaText

SETTEXT <TEXT>

Set the text to display into the MediaText area.

Example:

```
>SetText \"Tonight 21.30: Blue Diamond\"
<OK
```

SETBACKGROUNDCOLOR <A>,<R>,<G>,<B>

Set the MediaText background color supporting also the alpha channel

Example:

```
>SetBackgroundColor A=255,R=255,G=0,B=0
<OK
```

SETFOREGROUNDCOLOR <A>,<R>,<G>,<B>

Set the MediaText foreground color supporting also the alpha channel.

Example:

```
>SetForegroundColor A=255,R=255,G=0,B=0
<OK
```

SETFOREGROUNDCOLOR <A>,<R>,<G>,<B>

Set the MediaText foreground color supporting also the alpha channel.

Example:

```
>SetForegroundColor A=255,R=255,G=0,B=0
<OK
```

2.7 TextSlideShow

SETFILENAME <FILE>

Set the TextSlideShow file to process and display.

Example:

```
>SetFileName \"C://Dataource/plugIn.csv\"
<OK
```

2.8 SimpleMedia

SETFILENAME <FILE>,<LOOPS>,<HQ>,<FILLALPHA>,<USEALPHA>

Set the SimpleMedia filename to play and enable loop and other execution properties. The loop parameter is not required and its default value is false. Use HQ to resize the media with best quality.

The default SimpleMedia screen location is (0,0) and its size is the same of the loaded file.

Example:

```
>SetFileName \"C:\\\\Video\\video_loop.avi\",LOOP,HQ
<OK

>SetFileName \"C:\\\\Video\\video_loop.avi\",HQ
<OK
```

SETMEDIAPROPERTY <LOOP>, <HQ>, <FILLALPHA>, <USEALPHA>, <SCALETOFIT>, <FIELDPOLARITY>, <SYNCHTOVIDEOREFRESH>, <AUDIOONLY>, <AUDIODELAY>, <VOLUME>, <PAUSEFIRST>, <PAUSELAST>, <MARKINMILLIS>, <MARKOUTMILLIS>, <MARKIN>, <MARKOUT>, <MEDIATIMEFRAME>, <MEDIATIMEMILLIS>, <PIXELASPECTRATIO>

Set the SimpleMedia properties. You can set one or more properties in the same command:

LOOP play the media in loop mode [TRUE | FALSE];

HQ use the best quality when scaling the media [TRUE | FALSE];

FILLALPHA fill the alpha channel of the media [TRUE | FALSE];

USEALPHA compose the alpha channel of the media with the background [TRUE | FALSE];

SCALETOFIT scale the media to fit the size assigned to the plugin [TRUE | FALSE];

FIELDPOLARITY **(DEPRECATED)** set the media video fields polarity [0 = EVEN, 1 = ODD, 2 = NONE];

FIELDFLAGS set the media video fieldflags mode [0 = NONE, 1 = DEINTERLACE, 2 = EVEN_DOMINANT, 3 = ODD_DOMINANT];

SYNCHTOVIDEOREFRESH synchronize the media with the output device video refresh freq [TRUE | FALSE];

AUDIOONLY use only the audio of the media (no video) [TRUE | FALSE];

AUDIODELAY set A/V delay;

VOLUME [0..255] set the volume of the audio.

PAUSEFIRST enable or disable the pause at first frame [TRUE | FALSE];

PAUSELAST enable or disable the pause at last frame [TRUE | FALSE];

MARKINMILLIS set the mark in [The media time in milliseconds];

MARKOUTMILLIS set the mark out [The media time in milliseconds];

MARKIN set the mark in [The media time in frames];

MARKOUT set the mark out [The media time in frames];

MEDIATIMEFRAME set the media time [The media time in frames];

MEDIATIMEMILLIS set the media time [The media time in milliseconds];

PIXELASPECTRATIO set the pixel aspect ratio from preset or ratio in floating value

[

FREE,

SQUARE,

PAL_NTSC_4_3,

PAL_NTSC_16_9,

PAL_4_3_DV,

PAL_16_9_DV,

NTSC_4_3_DV,

NTSC_16_9_DV

]

Example:

```
>SetMediaProperty LOOP
<OK
```

enable the play loop

```
>SetMediaProperty LOOP=FALSE,VOLUME=180
<OK
```

disable play loop and set clip volume to 180

```
>SetMediaProperty LOOP, HQ, FILLALPHA, USEALPHA, SCALETOFIT, FIELDPOLARITY=1,
SYNCHTOVIDEOREFRESH, AUDIOONLY, AUDIODELAY=600, VOLUME=180
<OK

>SetMediaProperty  MARKIN=120
<OK

>SetMediaProperty  MARKOUT=200
<OK

>SetMediaProperty  MARKINMILLIS=4200
<OK

>SetMediaProperty  MARKOUTMILLIS=6200
<OK

>SetMediaProperty  FIELDFLAGS=0
<OK
```

GETMEDIAPROPERTY <LOOP>, <HQ>, <FILLALPHA>, <USEALPHA>, <SCALETOFIT>, <FIELDPOLARITY>, <SYNCHTOVIDEOREFRESH>, <AUDIOONLY>, <AUDIODELAY>, <VOLUME>, <PAUSEFIRST>, <PAUSELAST>, <MARKINMILLIS>, <MARKOUTMILLIS>, <MARKIN>, <MARKOUT>, <MEDIATIMEFRAME>, <MEDIATIMEMILLIS>, <PIXELASPECTRATIO>, <DURATION>, <TOTFRAMES>, <ALL>

Get the SimpleMedia properties. You can get one or more properties in the same command:

LOOP get media loop mode [TRUE | FALSE];

HQ get media best quality flag [TRUE | FALSE];

FILLALPHA get the fill alha channel flag [TRUE | FALSE];

USEALPHA get the compose the alpha channel flag of the media [TRUE | FALSE];

SCALETOFIT get the scale to fit flag of the media size assigned to the plugin [TRUE | FALSE];

FIELDFLAGS get the media video fieldflags mode [0 = NONE, 1 = DEINTERLACE, 2 = EVEN_DOMINANT, 3 = ODD_DOMINANT];

SYNCHTOVIDEOREFRESH get the synchronize the media with the output device video refresh freq flag [TRUE | FALSE];

AUDIOONLY get the use only the audio of the media flag [TRUE | FALSE];

AUDIODELAY get A/V delay;

VOLUME [0..255] get the volume of the audio.

PAUSEFIRST get the the pause at first frame status [TRUE | FALSE];

PAUSELAST get the pause at last frame status [TRUE | FALSE];

MARKINMILLIS get the mark in [The media time in milliseconds];

MARKOUTMILLIS get the mark out [The media time in milliseconds];

MARKIN get the mark in [The media time in frames, -1 if no set];

MARKOUT get the mark out [The media time in frames, -1 if no set];

MEDIATIMEFRAME get the current media time [The media time in frames];

MEDIATIMEMILLIS get the current media time [The media time in milliseconds];

PIXELASPECTRATIO get the pixel aspect ratio floating value

DURATION get the media duration [The media duration in milliseconds];

TOTFRAMES get the media duration in frames [The media duration in frames];

ALL retrieve all properties;

Examples:

```
>GETMEDIAPROPERTY FIELDFLAGS
<FIELDFLAGS=0

>GETMEDIAPROPERTY FIELDFLAGS, LOOP, HQ, FILLALPHA, USEALPHA, SCALETOFIT, SYNCHTOVIDEOREFRESH,
AUDIOONLY, AUDIODELAY, VOLUME, MARKIN, MARKOUT, MARKINMILLIS, MARKOUTMILLIS, MEDIATIMEFRAME,
MEDIATIMEMILLIS, DURATION, TOTFRAMES
<FIELDFLAGS=0,LOOP="FALSE",HQ="TRUE",FILLALPHA="FALSE",USEALPHA="FALSE",SCALETOFIT="FALSE",SYNC
HTOVIDEOREFRESH="TRUE",AUDIOONLY="FALSE",AUDIODELAY=200,VOLUME=255,MARKIN=0,MARKOUT=0,MARKINMIL
LIS=-1,MARKOUTMILLIS=-1,              MEDIATIMEFRAME=0,MEDIATIMEMILLIS=0,PIXELASPECTRATIO=8.333,
DURATION=1000,TOTFRAMES=25

>GETMEDIAPROPERTY ALL
<FIELDFLAGS=0,LOOP="FALSE",HQ="TRUE",FILLALPHA="FALSE",USEALPHA="FALSE",SCALETOFIT="FALSE",SYNC
HTOVIDEOREFRESH="TRUE",AUDIOONLY="FALSE",AUDIODELAY=200,VOLUME=255,MARKIN=0,MARKOUT=0,MARKINMIL
LIS=-1,MARKOUTMILLIS=-
1,MEDIATIMEFRAME=0,MEDIATIMEMILLIS=0,PIXELASPECTRATIO=8.333,DURATION=1000,TOTFRAMES=25
```

ADDFRAMEMARKER <FRAME>, <TYPE>, <LOOPS>

Add a FrameMarker at a specific frame.

FRAME frame number for adding the FrameMarker

TYPE FrameMarker type, it can assume one of the following values: TRIGGER_AND_PAUSE, TRIGGER_AND_CONTINUE, TRIGGER_AND_LOOP_START, TRIGGER_AND_LOOP_END, TRIGGER_AND_MARKIN, TRIGGER_AND_MARKOUT

LOOPS number of loops. It is valid only if the FrameMarker type is: TRIGGER_AND_LOOP_END

Examples:

```
>LBC_SENDCONTENTCOMMAND "SIGLA", "ADDFRAMEMARKER 185, TRIGGER_AND_CONTINUE"
<Ok

>LBC_SENDCONTENTCOMMAND "SIGLA", "ADDFRAMEMARKER 268, TRIGGER_AND_LOOP_END, 3"
<Ok
```

REMOVEFRAMEMARKER <FRAME>

Remove the FrameMarker at a specific frame.

FRAME frame number for deleting the FrameMarker

Examples:

```
>LBC_SENDCONTENTCOMMAND "SIGLA", "REMOVEFRAMEMARKER 185"
<Ok
```

GETFRAMEMARKERS

Return the list of the FrameMarker .

Examples:

```
>LBC_SENDCONTENTCOMMAND "SIGLA", "GETFRAMEMARKERS"
<FRAME=38,TYPE="TRIGGER_AND_MARKIN"
 FRAME=130,TYPE="TRIGGER_AND_LOOP_START"
 FRAME=240,TYPE="TRIGGER_AND_CONTINUE"
 FRAME=417,TYPE="TRIGGER_AND_PAUSE"
 FRAME=1583,TYPE="TRIGGER_AND_LOOP_END", LOOPS=3
```

```
FRAME=1696,TYPE="TRIGGER_AND_MARKOUT"
```

SETCROPINFO <ENABLE>, <PARAM0>, <PARAM1>, <PARAM2>, <PARAM3>

Set the crop info. You can set one or more properties in the same command:

ENABLE set the crop enable [TRUE | FALSE];

PARAM0
PARAM1
PARAM2
PARAM3 are the crop integer values in pixel;

This command respects the HTML#margins standard.

If you set 4 values are applied with order top/right/bottom/left
If you set 3 values are applied with order top/left/bottom
If you set 2 values the first value is applied to top/bottom and the second is applied to left/right
If you set 1 values are applied in all sides

Examples:

```
// ENABLE="TRUE",TOP=10,RIGHT=11,BOTTOM=12,LEFT=13
>SETCROPINFO ENABLE 10 11 12 13
<OK

// ENABLE="TRUE",TOP=10,RIGHT=0,BOTTOM=12,LEFT=13
>SETCROPINFO ENABLE 10 11 12
<OK

// ENABLE="TRUE",TOP=15,RIGHT=20,BOTTOM=15,LEFT=20
>SETCROPINFO ENABLE 15 20
<OK

// ENABLE="TRUE",TOP=10,RIGHT=10,BOTTOM=10,LEFT=10
>SETCROPINFO ENABLE 10
<OK
```

GETCROPINFO <ENABLE>, <TOP>, <RIGHT>, <BOTTOM>, <LEFT>, <ALL>

Set the crop info. You can set one or more properties in the same command:

ENABLE set the crop enable [TRUE | FALSE];

TOP get the crop top integer value in pixel;
RIGHT get the crop right integer value in pixel;
BOTTOM get the crop bottom integer value in pixel;
LEFT get the crop left integer value in pixel;

ALL retrieve all properties;

Examples:

```
>LBC_SENDCONTENTCOMMAND "SIGLA", "GETCROPINFO ENABLE TOP RIGHT BOTTOM LEFT"
<ENABLE="TRUE",TOP=10,RIGHT=0,BOTTOM=12,LEFT=11

>LBC_SENDCONTENTCOMMAND "SIGLA", "GETCROPINFO ALL"
<ENABLE="TRUE",TOP=10,RIGHT=0,BOTTOM=12,LEFT=11
```

2.9 Image

SETFILENAME <FILE>,<USEALPHA>,<HQ>

Set the Background filename to display. The USEALPHA parameter is not required and its default value is false. Use HQ to resize the media with best quality.

The default Background screen location is (0,0) and its size is the same of the loaded file.

Example:

```
>SetFileName \"C:\\\\Video\\video_loop.avi",ALPHA,HQ
<OK

>SetFileName \"C://Video/video_loop.avi",HQ
<OK
```

2.10 MediaPlayer

CLEARPLAYLIST

Clear the MediaPlayer files playlist.

Example:

```
>ClearPlaylist
<OK
```

ADDPLAYLISTFILE <FILEPATH>

Add the given file to the playlist at the last position. The file path may refer to either local or remote URL file.

Example:

```
>AddPlayListFile \"C://Datasource//tg.txt\"
<OK

>AddPlayListFile \"http://www.classx.it//prova.txt\"
<OK
```

RESETPLAYLIST

Reset the playlist execution forcing the MediaPlayer to restart by the first playlist file.

Example:

```
>ResetPlaylist
<OK
```

SETMEDIAPROPERTY <LOOP>, <HQ>, <FILLALPHA>, <USEALPHA>, <SCALETOFIT>, <FIELDPOLARITY>, <SYNCHTOVIDEOREFRESH>, <AUDIODELAY>, <VOLUME>

Set the SimpleMedia properties. You can set one or more properties in the same command:

LOOP play the media in loop mode;

HQ use the best quality when scaling the media;

FILLALPHA fill the alpha channel of the media;

USEALPHA compose the alpha channel of the media with the background;

SCALETOFIT scale the media to fit the size assigned to the plugin;

FIELDPOLARITY **(DEPRECATED)** set the media video fields polarity [0 = EVEN, 1 = ODD, 2 = NONE];

FIELDFLAGS set the media video fieldflags mode [0 = NONE, 1 = DEINTERLACE, 2 = EVEN_DOMINANT, 3 = ODD_DOMINANT];

SYNCHTOVIDEOREFRESH synchronize the media with the output device video refresh freq;

AUDIODELAY set A/V delay;

VOLUME [0..255] set the volume of the audio.

Example:

```
>SetMediaProperty LOOP
<OK
```

enable the play loop

```
>SetMediaProperty LOOP=FALSE,VOLUME=180
<OK
```

disable play loop and set clip volume to 180

```
>SetMediaProperty LOOP, HQ, FILLALPHA, USEALPHA, SCALETOFIT, FIELDPOLARITY=1,
SYNCHTOVIDEOREFRESH, AUDIODELAY=600, VOLUME=180
<OK
```

SETCROPINFO <ENABLE>, <PARAM0>, <PARAM1>, <PARAM2>, <PARAM3>

Set the crop info. You can set one or more properties in the same command:

ENABLE set the crop enable [TRUE | FALSE];

PARAM0
PARAM1
PARAM2
PARAM3 are the crop integer values in pixel;

This command respects the HTML#margins standard.

If you set 4 values are applied with order top/right/bottom/left
If you set 3 values are applied with order top/left/bottom
If you set 2 values the first value is applied to top/bottom and the second is applied to left/right
If you set 1 values are applied in all sides

Examples:

```
// ENABLE="TRUE",TOP=10,RIGHT=11,BOTTOM=12,LEFT=13
>SETCROPINFO ENABLE 10 11 12 13
<OK

// ENABLE="TRUE",TOP=10,RIGHT=0,BOTTOM=12,LEFT=13
>SETCROPINFO ENABLE 10 11 12
<OK

// ENABLE="TRUE",TOP=15,RIGHT=20,BOTTOM=15,LEFT=20
>SETCROPINFO ENABLE 15 20
<OK

// ENABLE="TRUE",TOP=10,RIGHT=10,BOTTOM=10,LEFT=10
>SETCROPINFO ENABLE 10
<OK
```

GETCROPINFO <ENABLE>, <TOP>, <RIGHT>, <BOTTOM>, <LEFT>, <ALL>

Set the crop info. You can set one or more properties in the same command:

ENABLE set the crop enable [TRUE | FALSE];

TOP get the crop top integer value in pixel;
RIGHT get the crop right integer value in pixel;
BOTTOM get the crop bottom integer value in pixel;
LEFT get the crop left integer value in pixel;

ALL retrieve all properties;

Examples:

```
>LBC_SENDCONTENTCOMMAND "SIGLA", "GETCROPINFO ENABLE TOP RIGHT BOTTOM LEFT"
<ENABLE="TRUE",TOP=10,RIGHT=0,BOTTOM=12,LEFT=11

>LBC_SENDCONTENTCOMMAND "SIGLA", "GETCROPINFO ALL"
<ENABLE="TRUE",TOP=10,RIGHT=0,BOTTOM=12,LEFT=11
```

2.11 SlideShow

CLEARPLAYLIST

Clear the Slideshow files playlist.

Example:

```
>ClearPlaylist
<OK
```

ADDPLAYLISTFILE <FILEPATH>

Add the given file to the playlist at the last position. The file path may refer to either local or remote URL file.

Example:

```
>AddPlayListFile \"C://Datasource//tg.txt\"
<OK

>AddPlayListFile \"http://www.classx.it//prova.txt\"
<OK
```

RESETPLAYLIST

Reset the playlist execution forcing the Slideshow to restart by the first playlist file.

Example:

```
>ResetPlaylist
<OK
```

2.12 NDI Source

SETCROPINFO <ENABLE>, <PARAM0>, <PARAM1>, <PARAM2>, <PARAM3>

Set the crop info. You can set one or more properties in the same command:

ENABLE set the crop enable [TRUE | FALSE];

PARAM0
PARAM1
PARAM2
PARAM3 are the crop integer values in pixel;

This command respects the HTML#margins standard.

If you set 4 values are applied with order top/right/bottom/left
If you set 3 values are applied with order top/left/bottom
If you set 2 values the first value is applied to top/bottom and the second is applied to left/right
If you set 1 values are applied in all sides

Examples:

```
// ENABLE="TRUE",TOP=10,RIGHT=11,BOTTOM=12,LEFT=13
>SETCROPINFO ENABLE 10 11 12 13
<OK
```

```
// ENABLE="TRUE",TOP=10,RIGHT=0,BOTTOM=12,LEFT=13
>SETCROPINFO ENABLE 10 11 12
<OK

// ENABLE="TRUE",TOP=15,RIGHT=20,BOTTOM=15,LEFT=20
>SETCROPINFO ENABLE 15 20
<OK

// ENABLE="TRUE",TOP=10,RIGHT=10,BOTTOM=10,LEFT=10
>SETCROPINFO ENABLE 10
<OK
```

GETCROPINFO <ENABLE>, <TOP>, <RIGHT>, <BOTTOM>, <LEFT>, <ALL>

Set the crop info. You can set one or more properties in the same command:

ENABLE set the crop enable [TRUE | FALSE];

TOP get the crop top integer value in pixel;
RIGHT get the crop right integer value in pixel;
BOTTOM get the crop bottom integer value in pixel;
LEFT get the crop left integer value in pixel;

ALL retrieve all properties;

Examples:

```
>LBC_SENDCONTENTCOMMAND "SIGLA", "GETCROPINFO ENABLE TOP RIGHT BOTTOM LEFT"
<ENABLE="TRUE",TOP=10,RIGHT=0,BOTTOM=12,LEFT=11

>LBC_SENDCONTENTCOMMAND "SIGLA", "GETCROPINFO ALL"
<ENABLE="TRUE",TOP=10,RIGHT=0,BOTTOM=12,LEFT=11
```

2.13 SharedFrame

SETCROPINFO <ENABLE>, <PARAM0>, <PARAM1>, <PARAM2>, <PARAM3>

Set the crop info. You can set one or more properties in the same command:

ENABLE set the crop enable [TRUE | FALSE];

PARAM0
PARAM1
PARAM2
PARAM3 are the crop integer values in pixel;

This command respects the HTML#margins standard.

If you set 4 values are applied with order top/right/bottom/left
If you set 3 values are applied with order top/left/bottom
If you set 2 values the first value is applied to top/bottom and the second is applied to left/right
If you set 1 values are applied in all sides

Examples:

```
// ENABLE="TRUE",TOP=10,RIGHT=11,BOTTOM=12,LEFT=13
>SETCROPINFO ENABLE 10 11 12 13
```

```
<OK

// ENABLE="TRUE",TOP=10,RIGHT=0,BOTTOM=12,LEFT=13
>SETCROPINFO ENABLE 10 11 12
<OK

// ENABLE="TRUE",TOP=15,RIGHT=20,BOTTOM=15,LEFT=20
>SETCROPINFO ENABLE 15 20
<OK

// ENABLE="TRUE",TOP=10,RIGHT=10,BOTTOM=10,LEFT=10
>SETCROPINFO ENABLE 10
<OK
```

GETCROPINFO <ENABLE>, <TOP>, <RIGHT>, <BOTTOM>, <LEFT>, <ALL>

Set the crop info. You can set one or more properties in the same command:

ENABLE set the crop enable [TRUE | FALSE];

TOP get the crop top integer value in pixel;
RIGHT get the crop right integer value in pixel;
BOTTOM get the crop bottom integer value in pixel;
LEFT get the crop left integer value in pixel;

ALL retrieve all properties;

Examples:

```
>LBC_SENDCONTENTCOMMAND "SIGLA", "GETCROPINFO ENABLE TOP RIGHT BOTTOM LEFT"
<ENABLE="TRUE",TOP=10,RIGHT=0,BOTTOM=12,LEFT=11

>LBC_SENDCONTENTCOMMAND "SIGLA", "GETCROPINFO ALL"
<ENABLE="TRUE",TOP=10,RIGHT=0,BOTTOM=12,LEFT=11
```

2.14 Video Input

SETSOURCEID <SOURCEID>

It changes the ID number of the source on the Video Input

SOURCEID ID number

Example:

```
>LBC_SENDCONTENTCOMMAND "contentName", "SetSourceID 2"
<OK
```

SETVOLUME <SETVOLUME>

Sets the volume to the specified video input source.

SETVOLUME percentage of volume

Example:

```
>LBC_SENDCONTENTCOMMAND "contentName", "SetVolume 10"
<OK
```

NB: this method does not modify in any way the master volume of the source. As soon as the master volume is reset, this volume is reset. (the values of the volume can be viewed on the tooltips of the slider of the VideoInput preview panel). This volume is relatively calculated compared to the master volume.

3 OutputDevice Remote Protocol

The Output device protocol collects a set of remote commands which allow to change some settings of the Output devices used by ClassX Applications. This document does not contain all remote protocol specifications and how to, but must be considered as an **add-on** to the main document.

This protocol changes the device settings whose names are collected into an enumeration named PARAMS. Any PARAM changes will have effect when the UPDATEFLDPARAMS command is called. Each parameter is described by a name that is unique.

The PARAMS supported by the protocol

```
FIELDMODE_PSF          // the psf video format

// PIP1 creation/update params
PIP1_VISIBLE           // 0 is invisible
PIP1_X                 // x coord
PIP1_Y                 // y coord
PIP1_WIDTH             // width
PIP1_HEIGHT            // height
PIP1_HIGHQUALITY       // 0 is low quality
PIP1_VOLUME            // audio volume [0 to 100]
PIP1_ALPHA             // alpha channel transparence [0 to 100]

PIP1_CROPLEFT          // left crop pixels
PIP1_CROPRIGHT         // right crop pixels
PIP1_CROPTOP           // top crop pixels
PIP1_CROPBOTTOM        // bottom crop pixels
PIP1_CROPENABLE        // crop enable (0 is disable)

// global pip params
PIP_OVERGRAPHICS       // 0 is under graphics (compositing), else is over graphics layer
PIP_ANIMSPEED          // 0 is immediate, else is the fps of the interpolation animation between
different PIP positions
PIP_ANIMRAMP           // 0 is motion linear, else is accelerated
```

FLD_GETPARAMS

Return a list of the device PARAMS. Each line of the list contains the parameter name and its current value in the format:

PARAM=<param>, VALUE=”<value>”.

Example:

```
-> FLD_GETPARAMS
<- PARAM=PIP_VISIBLE, VALUE="1"
PARAM=PIP0_X, VALUE="25"
PARAM=PIP0_Y, VALUE="65"
PARAM=PIP0_WIDTH, VALUE="39"
PARAM=PIP0_HEIGHT, VALUE="29"
PARAM=PIP0_HIGHQUALITY, VALUE="0"
```

FLD_GETPARAM[PARAM]

Return the current value of the given parameter.

Example:

```
-> FLD_GETPARAM PARAM=PIP_X
<- PARAM=PIP_X, VALUE="25"
```

FLD_SETPARAM[PARAM,VALUE]

Set the value into the parameter. Please remember that changes will be applied to the driver when the

UPDATEFLDPARAMS is called.

Example:

```
-> FLD_SETPARAM PARAM=PIP_Y VALUE="55"
<- OK
```

FLD_UPDATEPARAMS

Apply the current params values to the Output device. This call may require time when the device has to be reinitialized. Not at all params require the device reinitialization.

4 Error management

GETLASTERRORMESSAGE

Return the description of the last found error.

```
>GETLASTERRORMESSAGE
<UNKNOWN COMMAND
```

GETERRORMESSAGE <ERRORCODE>

Return the error message description according with the given ERRORCODE.

```
>ERRORMESSAGE 4
<INVALID FILE NAME
```

GETLASTERROR

Return the last found error.

```
>GETLASTERROR
<ERROR 2
```

SHOWDIALOGMESSAGES <ENABLE>

Disable warning or error messages displaying

```
>SHOWDIALOGMESSAGES ENABLE=FALSE
<OK
```

4.1 Error table

```
// Common errors
ERR_NONE                        =0,"No errors"
ERR_UNKNOWN                     =1,"Unknown error"
ERR_UNKNOWNCOMMAND              =2,"Unknown command"
ERR_INVALIDFILENAME             =3,"Invalid file name"
ERR_BADPARAMETERS               =4,"Bad parameters"
ERR_PLUGINCOMMANDFAILED         =5,"Plugin command failed"
ERR_NOTIMPLEMENTED              =6,"Command not implemented"
ERR_SHOULDQUIT                  =7,"Quit"
ERR_PARSE                       =8,"Command parsing error"
ERR_FILEDOESNTEXIST             =9,"The file does not exist"
ERR_FILEALREADYEXIST            =10,"The file already exist"

// CGTemplateEditor/CGPlayout Errors
CGPE_ERR_BASE                   =1000,"Error base for CGPlayout and CGTemplateEditor"
CGPE_ERR_LOADPROJECT            =1001,"Project load error"
CGPE_ERR_INVALIDENGINE          =1002,"Invalid engine"
CGPE_ERR_INVALIDINDEX           =1003,"Invalid project index"
CGPE_ERR_PROJECTNOTFOUND        =1004,"Project not found"
CGPE_ERR_COMMANDFORBIDDEN       =1005,"Command forbidden while playing"
CGPE_ERR_INVALIDFRAME           =1006,"Frame not valid"
```

```
CGPE_ERR_OBJECTNOTFOUND          =1007,"Object not found"
CGPE_ERR_NODATASOURCES           =1008,"No data sources available"
CGPE_ERR_NOLOADEDPROJECTS        =1009,"No loaded projects"
CGPE_ERR_MEDIAFILENOTFOUND       =1010,"Video or audio file not found"
CGPE_ERR_NOLOADEDPLUGINS         =1011,"No plugins loaded"
CGPE_ERR_NOEDITABLEOBJECTS       =1012,"No editable objects"
CGPE_ERR_INVALIDENGINEID         =1013,"Invalid engine ID, it is null or contains invalid chars or
                                        is already assigned"
CGPE_ERR_TIMEOUT                 =1014,"Timeout error waiting the plugin status"
CGPE_ERR_DEVICENOTINITIALIZED    =1015,"The device is not initialized"

// StealthCG errors
SCG_ERR_BASE                     =2000,"Error base for StealthCG"
SCG_ERR_INVALIDCONTENTID         =2001,"Invalid content ID, it is null or contains invalid chars"
SCG_ERR_CONTENTNOTFOUND          =2002,"Content not found"
SCG_ERR_PLUGINTYPE               =2003,"Unknown plugIn type"
SCG_ERR_INVALIDCONTENTOPERATION  =2004,"The operation is not supported by the content"
SCG_ERR_INVALIDTEMPLATEOBJECT    =2005,"Cannot find the template object to edit"
SCG_ERR_NOCONTENTCOMMANDS        =2006,"No commands for the content"
SCG_ERR_CONTENTALREADYEXISTS     =2007,"A content with the same ID already exists"
SCG_ERR_INVALIDTEMPLATE          =2008,"The template contains invalid data"
SCG_ERR_SAVEPROJECTFAILED        =2009,"Save project failed"
SCG_ERR_DEVICENOTINITIALIZED     =2010,"The device is not initialized"

// LBControl erros
LBC_ERR_BASE                     =3000,"Error base for LBControl"
LBC_ERR_CONTENTNOTFOUND          =3001,"Content not found"
LBC_ERR_INVALIDCONTENT           =3002,"Invalid content for the operation"
LBC_ERR_INVALIDTEMPLATEOBJECT    =3003,"Cannot find the template object to edit"
LBC_ERR_NOCONTENTCOMMANDS        =3004,"No commands for the content"
LBC_ERR_TEMPLATEOBJECTNOTFOUND   =3005,"Template object not found"

// Social Server errors
SCS_ERR_NOMESSAGES               =4000,"No messages"
```

5 Change log

Version 1.0.5

- [new] RECLAIMMEMORY implementation in order to cleanup unused runtime memory.
- [new] GETPROJECTINFO command returns new project properties: POSTERFRAME|FPS|RESOLUTION
- [mod] GETPROJECTINFO command returns a different frame marker description. The previous version returns "PAUSE" for the pause FrameMarker, while this version returns "TRIGGER_AND_PAUSE".
- [mod] GETPROJECTINFO command returns new FrameMarker types: TRIGGER_AND_CONTINUE, TRIGGER_AND_LOOP_START, TRIGGER_AND_LOOP_END.
- [fix] Fixed Take command that does not use the project posterframe when the take frame parameter is void.
- [new] Added GETPROTOCOLVERSION command that returns the protocol version.
- [new] Each project can be assigned to a PROJECTID, that is an unique string identifier. In the previous version the PROJECTID was given by the template position into a vector and its type was integer number. Now all commands that require the PROJECTID accept both types String and Integer, keeping the compatibility with previous version.
- [mod] The GETLOADEDPROJECTS command returns a list of elements where each item is rounded by quote char ' " '.
- [mod] GETSTATUS command returns PLAYER when a project is playing and also returns the project name and the playing time.

Version 1.0.6

- [new] Support OutputDeviceRemote protocol to control and setup the application output device

Index

1 How to use CGRemote protocol....2
1.1 Connecting to CGRemote....2
1.2 Special characters....2
1.3 Commands returning lists....3
1.4 Project identification....3
1.5 The Commands syntax parsing....3
1.6 Note....4
1.7 Application commands....5
Help....5
Log <source, text>....5
Load <cgp project path> [APPEND] [IGNORETEMPLATES] [IGNOREPLUGINS]....5
Quit <saveproject>....5
Clear....5
SetPlayMode [AUTO|LIVE]....5
GetPlayMode....5
SetOnAir [ON|OFF]....5
GetOnAir....5
GetStatus [PLAYER|LOADER]....6
GetLoadedProjects....6
GetInfo <PROJECTFILENAME | RESOLUTION, PROJECTDIR>....6
RestartGFBServer....6
ReclaimMemory....6
1.8 Graphics templates commands....7
AddProject <GFXPROJECTPATH, PROJECTID>....7
DelProject <GFXPROJECTPATH>....7
DelProjectByID <POSID>....7
SelectProjectByID <POSID>....7
SelectProject <GFXPROJECTPATH>....7
SetProjectSpeed <SPEEDPERC>....7
SetProjectLoops <LOOPS>....7
AddMediaProject <VIDEOFILE=videomediafile [AUDIOFILE=audiomediafile]>....7
SetProjectMediaFile <VIDEOFILE=videomediafile [AUDIOFILE=audiomediafile]>....7
SetProjectMediaProperties <[
VIDEOFILE_ALPHAENABLE=[true | false] VIDEOFILE_ALPHAFILLED=[true | false]
VIDEOFILE_AUDIODELAY=[0..4000]
VIDEOFILE_AUDIOONLY=[true | false]
VIDEOFILE_FIELDSPOLARITY=[EVEN | ODD | NONE]
VIDEOFILE_LOOP=[true | false]
VIDEOFILE_SCALEHQ=[true | false]
VIDEOFILE_SCALETOFIT=[true | false]
VIDEOFILE_SYNCHTOVIDEOREFRESH=[true | false]
VIDEOFILE_VOLUME=[0..255]
AUDIOFILE_LOOP=[true | false]
]>....8
UpdateDataSources....8
ReloadProjectDataSourcesByID <PROJECTID>....8
ReloadProjectDataSources <PROJECTNAME>....8
Start....8
Stop....8
Take [<frame>,<takemode[SINGLE|START_SEQUENCE|END_SEQUENCE]>]....8
TakeAtRatio [<currenttime>,<totaltime>,<takemode[SINGLE|START_SEQUENCE|
END_SEQUENCE]>]....9
OpenProjectEditor 9
1.9 Editable objects of a graphic template....10
GetProjectObjects....10
GetProjectInfo <TOTFRAMES | NOBJECTS | CURRFRAME | DESCRIPTION | AUDIOFILE |
VIDEOFILE | GFXFILE | ICONFILE|FRAMEMARKER|POSTERFRAME|FPS|RESOLUTION>....10
EditProjectObject <OBJECTNAME>,<VALUE>....11
1.10 MediaPlugs commands....11

GetAvailablePlugins 11
GetPluginStatus <PLUGINID> 11
SendPluginCommand <PLUGINID>,<PLUGINCOMMAND> 12
StartPlugin <pluginID> 12
StopPlugin <pluginID> 12
StopAllPlugins 12
SetPluginBounds <pluginID>,<X>,<Y>,<WIDTH>,<HEIGHT> 12
WaitPluginStatus <pluginID>, <RUNNING | SUSPENDED | STOPPED>, [<timeout>] 12
2 MediaPlug commands 13
3 Introduction 1
4 The MediaPlug 2
4.1 The playlist 2
4.2 Automatic Mode 3
4.2.1Playlist from folder 3
4.2.2Playlist from file 3
4.2.3Advanced options for Ticker and StyledTicker 3
4.3 Background 4
4.4 DigiClock 5
4.5 Clock 6
4.6 Logo 7
4.7 MediaPlayer 8
4.7.1.1MediaCapture 10
4.8 SlideShow 12
4.9 Ticker Horizontal 15
4.10 SMSTicker 16
4.11 Karaoke 18
4.11.1Manual Mode 18
4.12 MediaText 20
4.13 ScreenCapture 21
4.14 MediaShape 22
4.15 LiveTimer 22
4.16 TextShow 23
4.17 SharedFrame 25
4.17.1Create a SharedFrame plugin 25
4.17.2Use a SharedFrame plugin 26
4.18 StyledTicker 27
4.18.1The “Appearance” panel 28
4.18.2The “HTML Settings” panel 28
4.18.3The “Execution time” section 29
4.19 Text Object 30
4.19.1Filling of an object 32
4.19.1.1Solid Filling 32
4.19.1.2Filling and radial gradient 32
4.19.1.3Fill options 32
4.19.1.4Overview of the fill 33
4.19.1.5Presets 33
4.19.2Edge of a text 33
4.19.2.1Edge type line 34
4.19.2.2Edge dashed type 34
4.19.2.3Edge wade type 34
4.19.2.4Edge text type 34
4.19.3Support HTML tag 35
4.20 Shape Object 36
4.20.1“Rectangle” Shape 36
4.20.2“Ellipse” V Shape 37
4.20.3“Polygon” Shape 37
4.20.4“General path” Shape 37
4.20.5Shape “chart” 38
4.20.6Pie Chart 38
4.20.7Hystogram chart 38
4.20.8Line chart 39

4.20.9Filling the shape....39
4.20.10Border and padding on the edge shape....39
4.21 Connect object values with external DataSource....40
5 Place of....41
6 For user....43
6.1 Frequently Asked Questions....43
7 OutputDevice Remote Protocol....46
The PARAMS supported by the protocol....46
FLD_GETPARAMS....46
FLD_GETPARAM[PARAM]....47
FLD_SETPARAM[PARAM,VALUE]....47
FLD_UPDATEPARAMS....47
8 Error management....48
GETLASTERRORMESSAGE....48
GETERRORMESSAGE <ERRORCODE>....48
GETLASTERROR....48
SHOWDIALOGMESSAGES <ENABLE>....48
8.1 Error table....48
9 Change log....49